

LOG ROTATION ON AWS EC2

Full Step-by-Step Practical Guide

PART 1 — Connect & Inspect

```
# Connect to EC2
ssh -i "your-key.pem" ec2-user@your-ec2-public-ip # Amazon Linux
ssh -i "your-key.pem" ubuntu@your-ec2-public-ip   # Ubuntu

# Check logrotate version
logrotate --version

# View existing configs
cat /etc/logrotate.conf
ls -la /etc/logrotate.d/

# Check disk usage – CRITICAL: default EC2 root is only 8GB!
du -sh /var/log/*
df -h
```

PART 2 — Install Nginx & Generate Logs

```
# Amazon Linux 2
sudo yum update -y && sudo yum install nginx -y
sudo systemctl start nginx && sudo systemctl enable nginx

# Ubuntu
sudo apt update && sudo apt install nginx -y

# Generate access logs
curl http://localhost
curl http://localhost/notfound # Generates 404s

# Watch live
sudo tail -f /var/log/nginx/access.log
ls -lh /var/log/nginx/
```

PART 3 — Custom App Log Rotation

Step 1 — Simulate App Writing Logs

```
sudo mkdir -p /var/log/myapp
sudo chown ec2-user:ec2-user /var/log/myapp

cat > /home/ec2-user/simulate_logs.sh << 'EOF'
#!/bin/bash
while true; do
    echo "$(date '+%Y-%m-%d %H:%M:%S') INFO User login: user_${(( RANDOM % 1000 ))}" >> /var/log/myapp/app.log
    echo "$(date '+%Y-%m-%d %H:%M:%S') ERROR DB timeout: ${(( RANDOM % 5000 ))}ms" >> /var/log/myapp/app.log
    sleep 1
done
```

```
done
EOF
chmod +x /home/ec2-user/simulate_logs.sh
nohup /home/ec2-user/simulate_logs.sh &
tail -f /var/log/myapp/app.log
```

Step 2 — Create logrotate Config

```
sudo nano /etc/logrotate.d/myapp

/var/log/myapp/*.log {
    daily
    rotate 7
    compress
    delaycompress
    missingok
    notifempty
    create 0644 ec2-user ec2-user
    dateext
    dateformat -%Y-%m-%d
    size 10M
    postrotate
        echo "Log rotated at $(date)" >> /var/log/myapp/rotation.log
    endscript
}
```

Step 3 — Test & Force Rotation

```
# Dry run – no actual changes
sudo logrotate --debug /etc/logrotate.d/myapp

# Force rotation NOW
sudo logrotate --force /etc/logrotate.d/myapp

# Verify
ls -lh /var/log/myapp/
# app.log                current (new empty file)
# app.log-2026-05-08     yesterday's (uncompressed)
# app.log-2026-05-07.gz  older logs compressed
```

PART 4 — Automate with Cron

```
# Check existing logrotate cron
cat /etc/cron.daily/logrotate

# Verify cron service
sudo systemctl status crond      # Amazon Linux
sudo systemctl status cron       # Ubuntu

# Add hourly rotation for high-traffic
sudo crontab -e
# Add:
0 * * * * /usr/sbin/logrotate /etc/logrotate.d/myapp > /var/log/myapp/cron-rotate.log 2>&1
```

PART 5 — Ship Logs to S3

```
# Create S3 bucket
aws s3 mb s3://my-ec2-logs-backup-$(date +%Y%m%d)
aws s3api put-bucket-versioning \
  --bucket my-ec2-logs-backup \
  --versioning-configuration Status=Enabled
```

IAM Role Policy (attach to EC2 instance)

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:PutObject", "s3:GetObject", "s3:ListBucket"],
    "Resource": [
      "arn:aws:s3:::my-ec2-logs-backup",
      "arn:aws:s3:::my-ec2-logs-backup/*"
    ]
  }]
}
```

Auto-Upload in postrotate

```
postrotate
  ROTATED="/var/log/myapp/app.log-$(date --date='yesterday' +%Y-%m-%d).gz"
  if [ -f "$ROTATED" ]; then
    aws s3 cp "$ROTATED" \
      s3://my-ec2-logs-backup/$(hostname)/$(date +%Y/%m)/ \
      --storage-class STANDARD_IA
    echo "$(date): Uploaded $ROTATED to S3" >> /var/log/myapp/s3-upload.log
  fi
endscript
```

PART 6 — CloudWatch Integration

```
# Install CloudWatch Agent
sudo yum install amazon-cloudwatch-agent -y      # Amazon Linux

{
  "logs": {
    "logs_collected": {
      "files": {
```

```
        "collect_list": [
            {
                "file_path": "/var/log/myapp/app.log",
                "log_group_name": "/ec2/myapp/application",
                "log_stream_name": "{instance_id}"
            },
            {
                "file_path": "/var/log/nginx/access.log",
                "log_group_name": "/ec2/nginx/access",
                "log_stream_name": "{instance_id}"
            }
        ]
    }
}

sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
    -a fetch-config -m ec2 \
    -c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -s

sudo systemctl status amazon-cloudwatch-agent
```

PART 7 — Disk Space Alert Script

```
cat > /home/ec2-user/disk_alert.sh << 'EOF'
#!/bin/bash
THRESHOLD=80
USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')
if [ "$USAGE" -gt "$THRESHOLD" ]; then
    echo "ALERT: EC2 disk usage is ${USAGE}% on $(hostname)" | \
    aws sns publish \
        --topic-arn arn:aws:sns:ap-south-1:123456789:disk-alerts \
        --message file:///dev/stdin \
        --subject "EC2 Disk Alert - $(hostname)"
fi
EOF
chmod +x /home/ec2-user/disk_alert.sh

# Run every 15 min
crontab -e
# Add: */15 * * * * /home/ec2-user/disk_alert.sh
```

Quick Reference Cheatsheet

```
# Test config (dry run)
sudo logrotate --debug /etc/logrotate.d/myapp

# Force rotate now
sudo logrotate --force /etc/logrotate.d/myapp

# Check rotation history
sudo cat /var/lib/logrotate/logrotate.status

# Check disk usage
```

```
df -h && du -sh /var/log/*

# Find large log files
find /var/log -name "*.log" -size +100M

# Manually compress old logs
gzip /var/log/myapp/app.log.1
```

Full Architecture Overview

Component	Role
App	Writes logs to /var/log/myapp/app.log
logrotate (cron)	Rotates, compresses, triggers postrotate script
postrotate script	Uploads rotated .gz files to S3 STANDARD_IA
CloudWatch Agent	Streams live logs to CloudWatch Log Groups
S3 Lifecycle Policy	Auto-moves to Glacier after 90 days
SNS Alarm	Alerts when disk usage exceeds 80%